

Jack Bajcz

Graduate Research Engineer | Cryogenic Systems & Thermal Analysis | EIT

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Education

Michigan State University, MS in Mechanical Engineering – East Lansing, MI Jan 2025 – present

- GPA: 3.937/4.0
- Tentative Thesis Title: Performance Characterization of a Sub-ppmv Level Freeze-out Purification System for Helium

Michigan State University, BS in Mechanical Engineering – East Lansing, MI Aug 2020 – Dec 2024

- Degree Concentration in Computational Design
- Minors: Computer Science and German

Experience

Graduate Research Assistant, Facility for Rare Isotope Beams – East Lansing, MI Jan 2025 – present

- Conducting original thesis research on the transient thermal performance of a novel coiled finned tube heat exchanger in a freeze-out helium purification system, achieving sub-ppmv purity to protect FRIB's cryogenic cold box from frost-induced damage to heat transfer and turbine surfaces.
- Developing analytical thermal models correlating temperature, pressure, and flow sensor data across the purification skid to characterize frost deposition location (shell-side vs. heat transfer surfaces) and its effect on heat exchanger performance degradation over time.
- Benchmarking in-house purifier performance against a commercially available freeze-out system, investigating the thermal and operational tradeoffs of finned tube geometry in freeze-out applications.
- Engineering CAD assemblies, fabrication drawings, and P&IDs for the cryogenic purification system and supporting infrastructure in SolidWorks, including adapting the humid gas generator interface for experimental testing.

Student Technical Assistant I, Facility for Rare Isotope Beams – East Lansing, MI Feb 2023 – Dec 2024

- Supported CAD modeling and documentation of cryogenic transfer line systems in SolidWorks under senior engineering guidance.
- Assisted in P&ID drafting and review, developing familiarity with ASME B31.3 piping code requirements.
- Developed internal tools to reduce engineering time spent on repetitive tasks by 30% using VBA.

Engineering and 3D Printing Intern, The Hatch - Burgess Institute, MSU – East Lansing, MI Mar 2023 – May 2024

- Managed a fleet of 3D printers, implementing hardware upgrades that reduced downtime by 50%.
- Provided engineering consultation and support for 10+ student projects.

Drafting Intern, Facilities & Operations, University of Michigan – Ann Arbor, MI July 2022 – Aug 2022

- Updated detailed Building Information Models (BIMs) of existing systems using AutoCAD.
- Completed laboratory and roof inspections of 30 campus research facilities, ensuring safety requirements.

Project Support Engineer, Air Force Institute of Technology – Wright-Patterson AFB, OH May 2021 – Aug 2021

- Programmed data collection and analysis tools in Matlab and Python, accelerating academic research.
- Successfully completed extensive background check and was granted secret clearance (inactive).

Resident Assistant, Michigan State University – East Lansing, MI

Jan 2022 – present

- Supervised a residential floor of ~30 students, providing guidance, conflict mediation, and crisis response.
- Coordinated community programming and collaborated with university housing staff on policy enforcement.

Presentations

Performance Characterization of a Sub-ppmv Level Freeze-out Purification System for Helium – MSU Council of Graduate Students Graduate Academic Conference, East Lansing, MI 2026

Poster presentation on thesis research.

Performance Characterization of a Sub-ppmv Level Freeze-out Purification System for Helium – International Cryogenic Engineering Conference (ICEC), Daejeon, South Korea 2026

Accepted oral presentation — forthcoming.

Projects

Heat Exchanger Surrogate Model – ASME IDETC/CIE Student Hackathon Aug 2025

A machine learning model to predict heat exchanger performance metrics without numerical simulation.

- Written in Python, using Tensorflow and Keras.
- 3rd Place Overall in category
- Optimized solution for consumer-level hardware to augment dataset and run inference.

Liquid Cooling System Steady State Design Tool – Eaton Aerospace Sept 2024 – Dec 2024

Degree capstone project to design a steady-state analysis tool for liquid cooling systems in aircraft applications.

- Developed a highly extensible MATLAB code library of components and a system-level solver.
- Successfully presented design to Eaton Aerospace engineers and faculty advisors.

Skills

Analysis & Modeling: Large-Scale Cryogenic Systems, Heat Exchangers, Fluid Dynamics, Thermal Systems

CAD & Software: Solidworks, Autodesk Inventor, ANSYS Mechanical, ANSYS Fluent, nTop, Microsoft Office

Programming: Python, MATLAB, LaTeX, VBA, C++

Certificates and Certifications

Engineer In Training (EIT), Tennessee & New York, 2026

FE - Fundamentals of Engineering, NCEES, 2025

NQA-1 Part 1 - 18 QA Requirements, ASME, 2026

Accelerator Science and Engineering Traineeship in Large-Scale Cryogenic Systems, FRIB/MSU, *In Progress*

Graduate Certificate in Computational Modeling, MSU, *In Progress*

Certificate in Drawing Interpretation (GD&T), ASME, 2026

Certificate in Additive Manufacturing (LPBF), ASME, 2026

Professional Organizations

Cryogenic Society of America (CSA)

American Society of Mechanical Engineers (ASME)

Institute of Electrical and Electronics Engineers (IEEE)

National Society of Professional Engineers (NSPE)

Society of Manufacturing Engineers (SME)